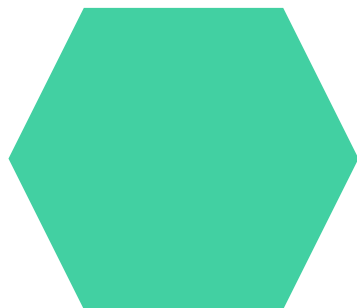
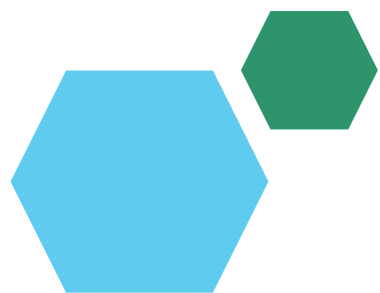


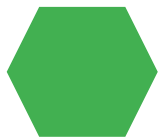


Project Title -Seasonal Agriculture Performance Analysis

[Student Name - Nishkarsh Antil

College Name - HMR Institute of technology]

AICTE STU ID : STU6a67024e424331785135694



PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description



->Dataset includes farming practices, environmental conditions, crop production, resource usage, and economic performance.

->Analysis focuses on **seasonal differences** in agricultural outcomes.

Techniques: **Data cleaning, grouping, correlation, visualization.**



WHO ARE THE END USERS?

- 
- **Farmers** - better planning for seasonal crops.
 - **Policy Makers** - evidence-based agricultural policies.
 - **Researchers/Students** - real-world data analytics practice.
 - **Agri-businesses** - optimize resource usage and market strategies.
- 
- 

Technology Used

- Python (Pandas, NumPy) - data cleaning & analysis
- Matplotlib, Seaborn - visualization
- Google Colab - cloud-based Python environment
- GitHub - version control & sharing



RESULTS

Data types:

```
Farm_ID      object
State        object
District     object
Crop         object
Season       object
Farm_Area_Hectares float64
Rainfall_mm  float64
Avg_Temperature_C float64
Humidity_pct  float64
Sunlight_Hours_Day float64
Soil_pH       float64
Soil_Moisture_pct float64
Nitrogen_kg_ha float64
Phosphorus_kg_ha float64
Potassium_kg_ha float64
Irrigation_Method object
Fertilizer_kg_ha float64
Pesticide_Litre_ha float64
Seed_Quality_Score float64
Yield_Tonnes_Ha float64
Production_Tonnes float64
Market_Price_INR_Tonne int64
Total_Cost_INR int64
Revenue_INR int64
Profit_INR int64
Water_Used_m3 int64
Water_Efficiency_t_per_1000m3 float64
Disease_Pest_Risk_pct float64
dtype: object
```

Missing values:

```
Farm_ID      0
State        0
District     0
Crop         0
Season       0
Farm_Area_Hectares 0
Rainfall_mm  48
Avg_Temperature_C 0
Humidity_pct  0
Sunlight_Hours_Day 0
Soil_pH       0
Soil_Moisture_pct 40
Nitrogen_kg_ha 0
Phosphorus_kg_ha 0
Potassium_kg_ha 0
Irrigation_Method 0
Fertilizer_kg_ha 0
Pesticide_Litre_ha 0
Seed_Quality_Score 0
Yield_Tonnes_Ha 32
Production_Tonnes 0
Market_Price_INR_Tonne 0
Total_Cost_INR 0
Revenue_INR 0
Profit_INR 0
Water_Used_m3 0
Water_Efficiency_t_per_1000m3 0
Disease_Pest_Risk_pct 0
dtype: int64
```

RESULTS

Statistical summary:

	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct \
count	3880.000000	3880.000000	3880.000000	3880.000000
mean	7.878059	601.676804	26.811031	63.279639
std	4.222548	288.326449	3.809380	11.970706
min	0.500000	80.000000	16.200000	25.000000
25%	4.250000	370.975000	24.000000	54.900000
50%	7.880000	582.900000	26.900000	63.100000
75%	11.600000	834.600000	29.600000	72.000000
max	15.000000	1395.200000	39.700000	95.000000

	Sunlight_Hours_Day	Soil_ph	Soil_Moisture_pct	Nitrogen_kg_ha \
count	3880.000000	3880.000000	3880.000000	3880.000000
mean	7.319072	6.701521	26.541211	120.165876
std	1.124002	0.640693	6.704632	31.268146
min	3.500000	5.200000	8.000000	40.000000
25%	6.600000	6.260000	21.800000	98.400000
50%	7.300000	6.690000	26.500000	119.900000
75%	8.100000	7.130000	31.300000	141.200000
max	11.000000	8.400000	47.000000	220.000000

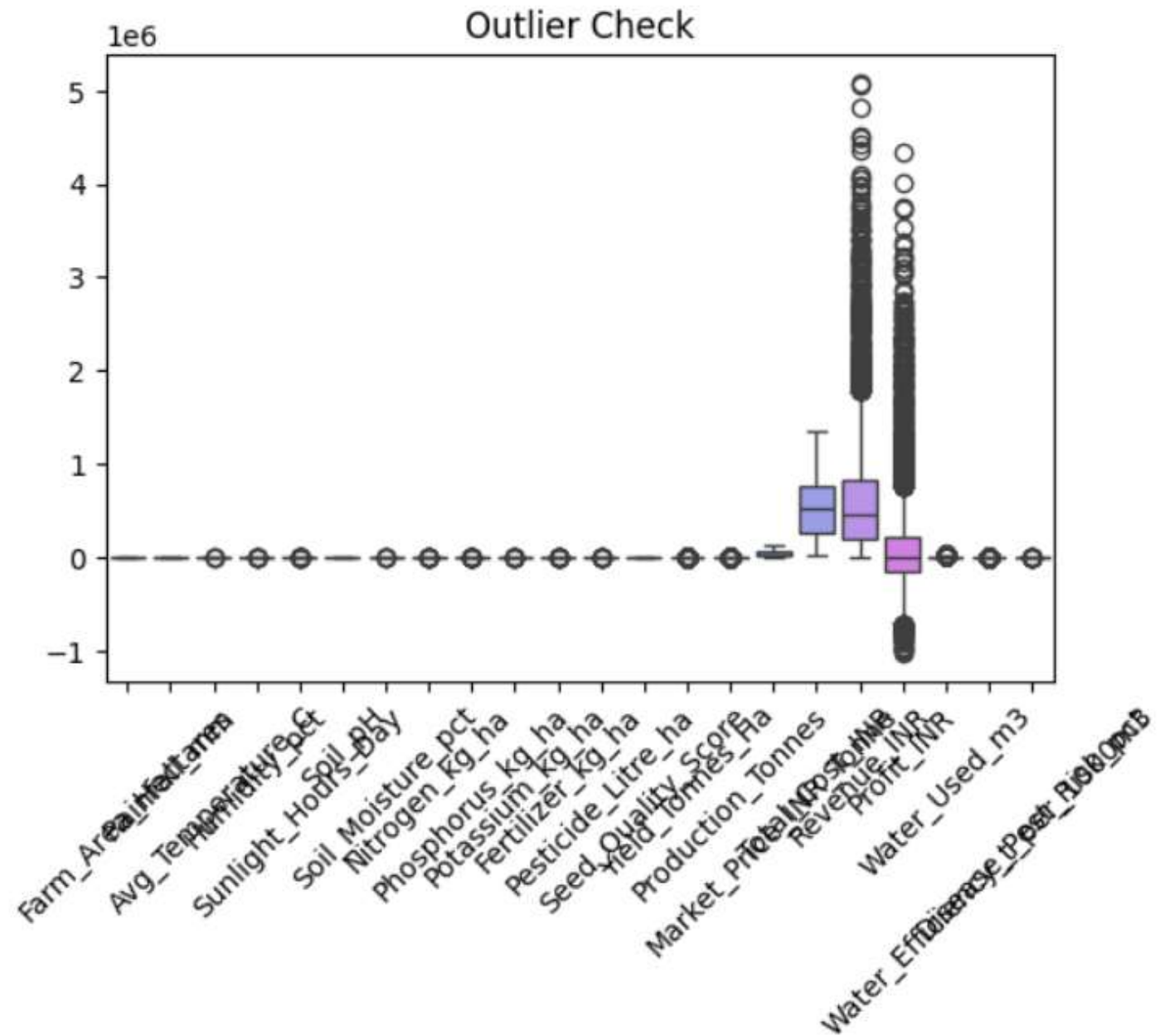
	Phosphorus_kg_ha	Potassium_kg_ha	...	Seed_Quality_Score \
count	3880.000000	3880.000000	...	3880.000000
mean	57.790851	104.631856	...	0.826031
std	17.014782	27.676885	...	0.101858
min	15.000000	30.000000	...	0.650000
25%	46.275000	86.100000	...	0.730000
50%	57.500000	105.100000	...	0.830000
75%	69.400000	123.500000	...	0.910000
max	120.000000	200.000000	...	1.000000

	Yield_Tonnes_Ha	Production_Tonnes	Market_Price_INR_Tonne \
count	3880.000000	3880.000000	3880.000000
mean	5.241152	43.082593	44847.972423
std	13.315214	127.043309	30251.187889
min	0.300000	0.190000	2594.000000
25%	1.047500	5.020000	21461.500000
50%	1.730000	11.730000	25956.500000
75%	2.830000	24.305000	68339.500000
max	101.440000	1424.400000	131240.000000

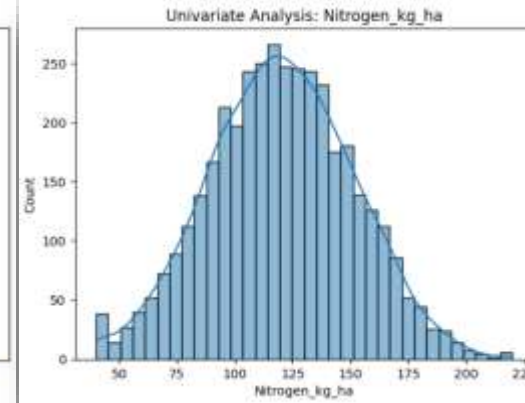
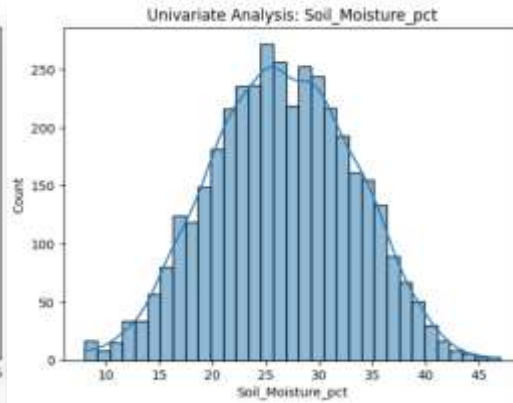
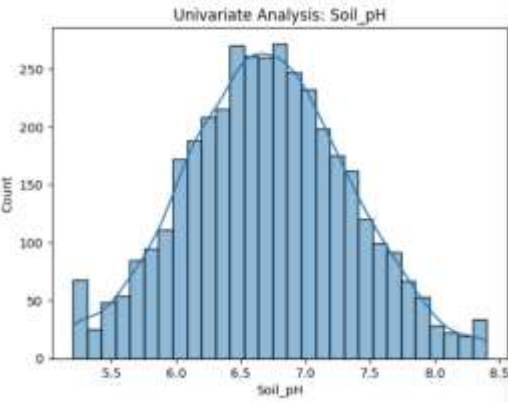
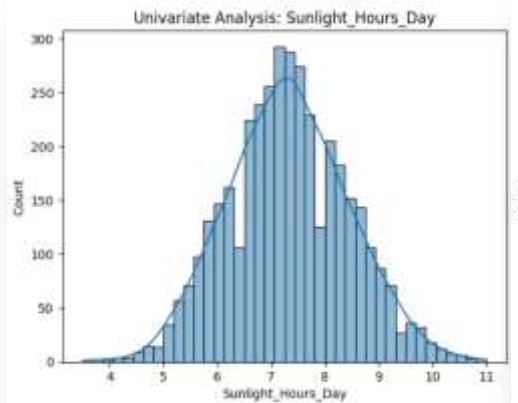
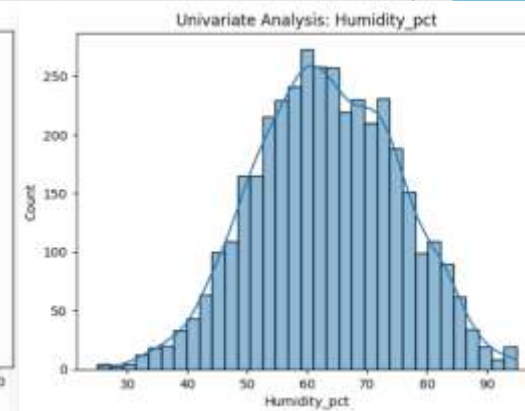
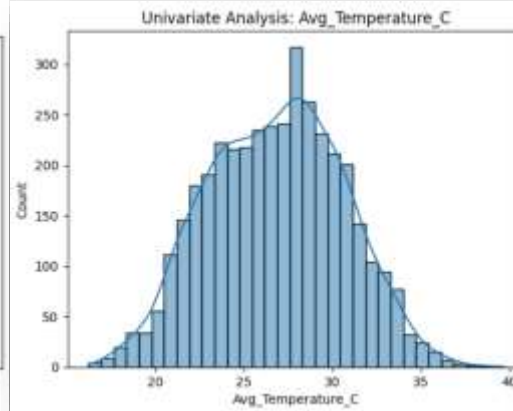
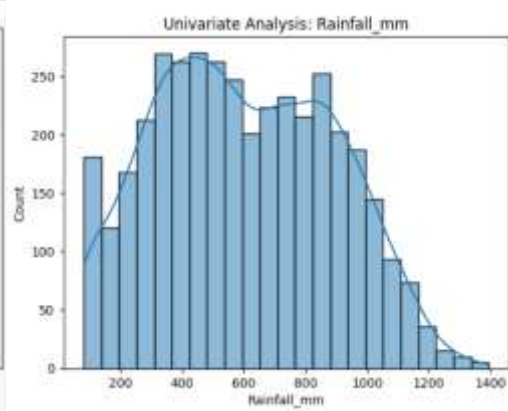
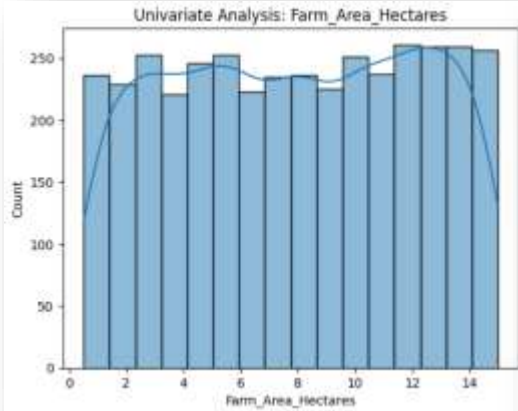
	Total_Cost_INR	Revenue_INR	Profit_INR	Water_Used_m3 \
count	3.880000e+03	3.880000e+03	3.880000e+03	3880.000000
mean	5.253930e+05	6.376869e+05	1.122939e+05	6009.920876
std	2.941896e+05	6.381082e+05	5.465619e+05	5548.730467
min	2.682600e+04	6.107000e+03	-1.019223e+06	128.000000
25%	2.730532e+05	2.096812e+05	-1.486638e+05	2251.000000
50%	5.158075e+05	4.564110e+05	4.039000e+03	4408.500000
75%	7.618188e+05	8.322630e+05	2.164538e+05	7787.250000
max	1.349990e+06	5.080900e+06	4.345421e+06	40902.000000

	Water_Efficiency_t_per_1000m3	Disease_Pest_Risk_pct
count	3880.000000	3880.000000
mean	5.389393	46.396108
std	8.854635	12.449748
min	0.135000	5.000000
25%	1.660750	37.300000
50%	2.951500	46.400000
75%	5.158000	55.000000
max	79.800000	88.900000

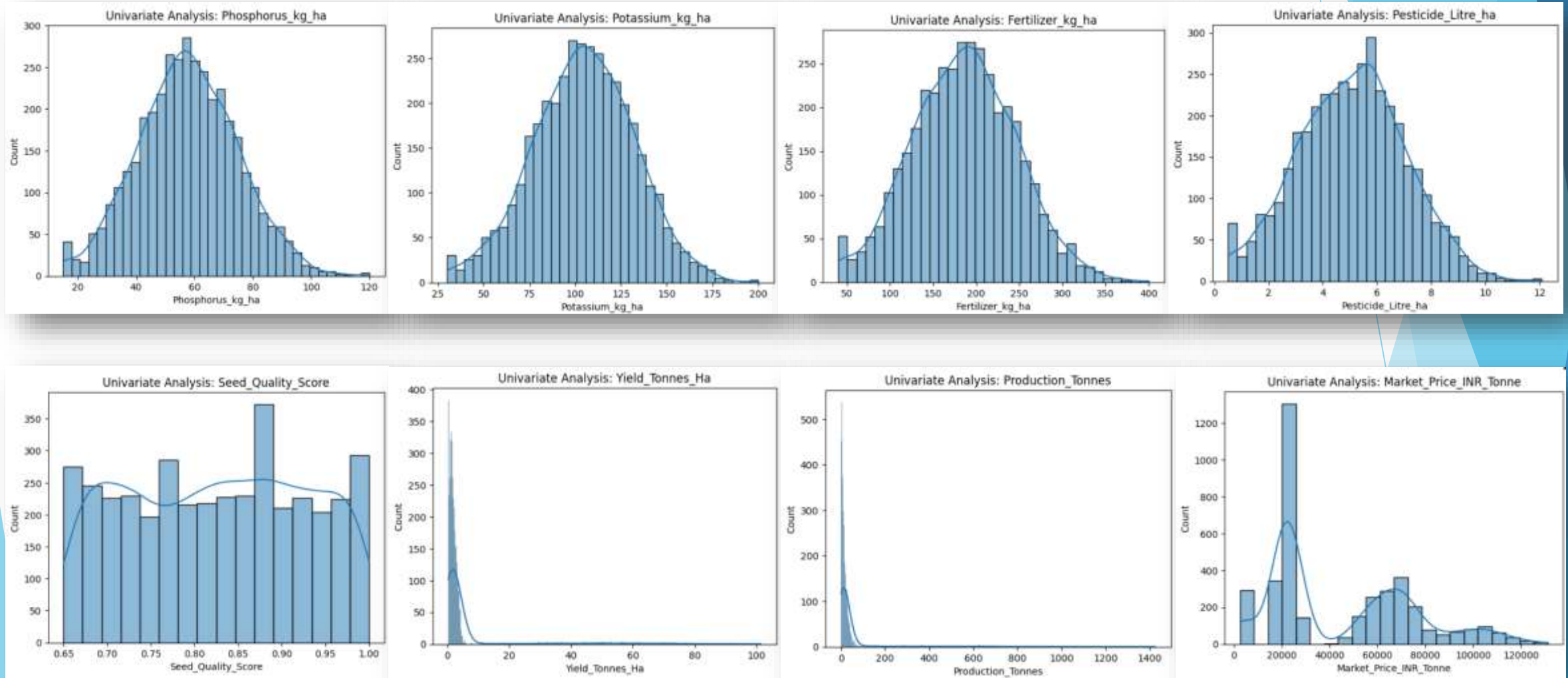
RESULTS



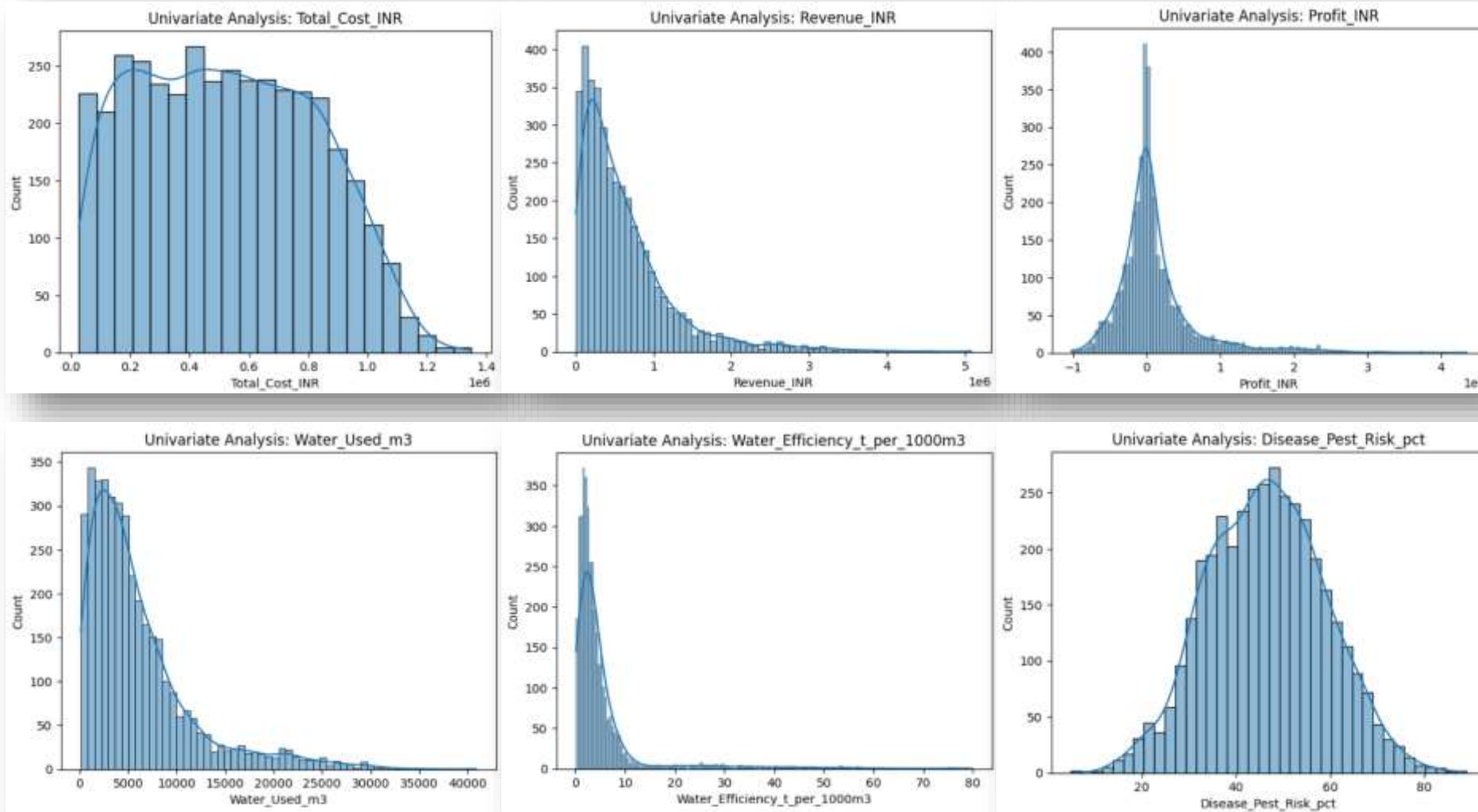
RESULTS



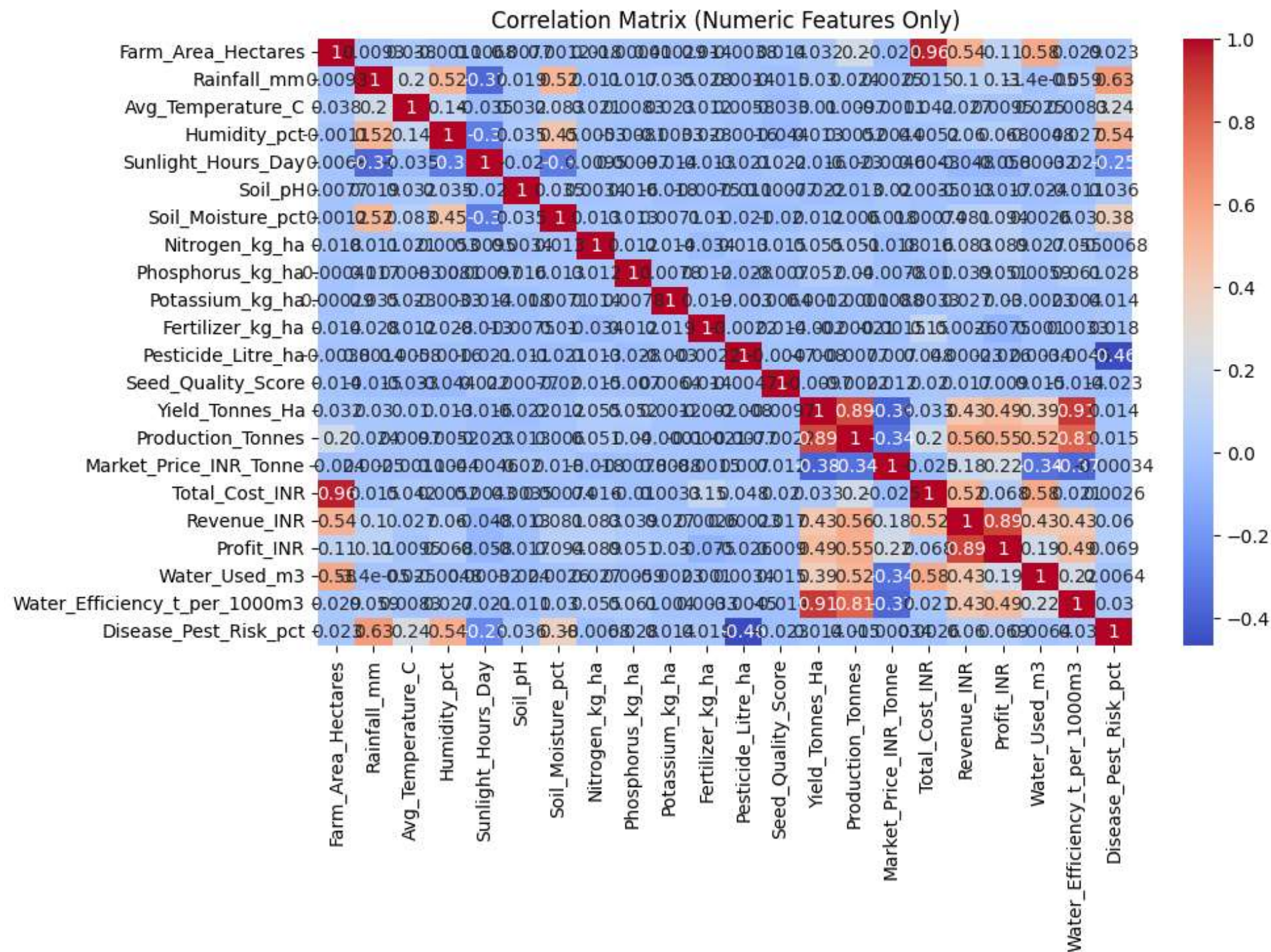
RESULTS



RESULTS

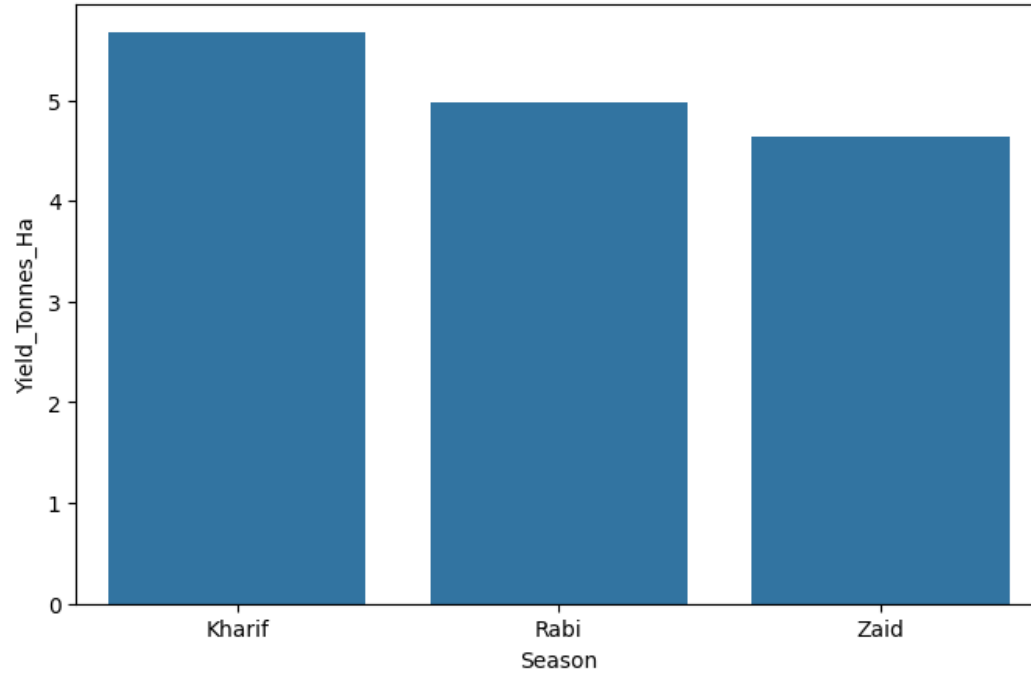


RESULTS

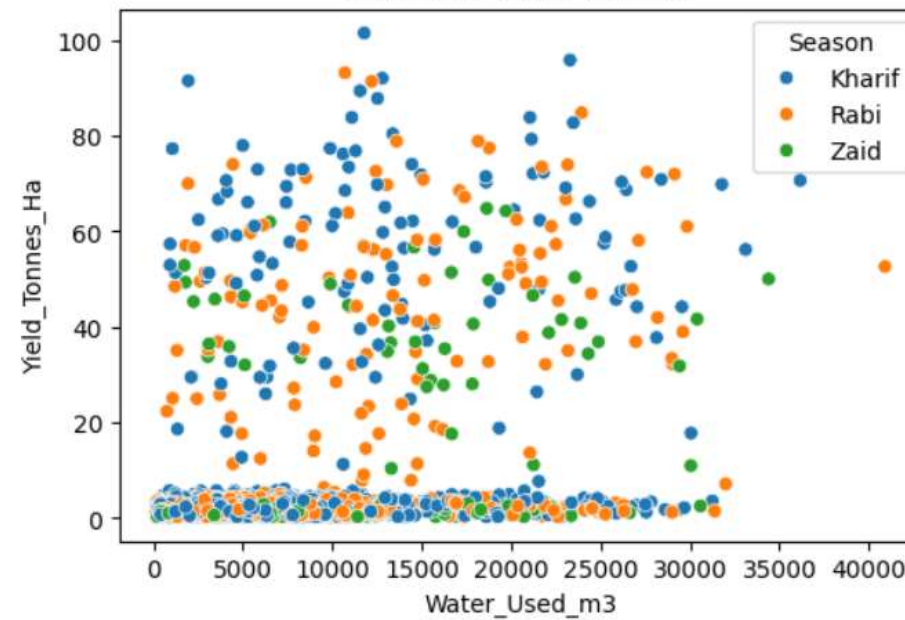


RESULTS

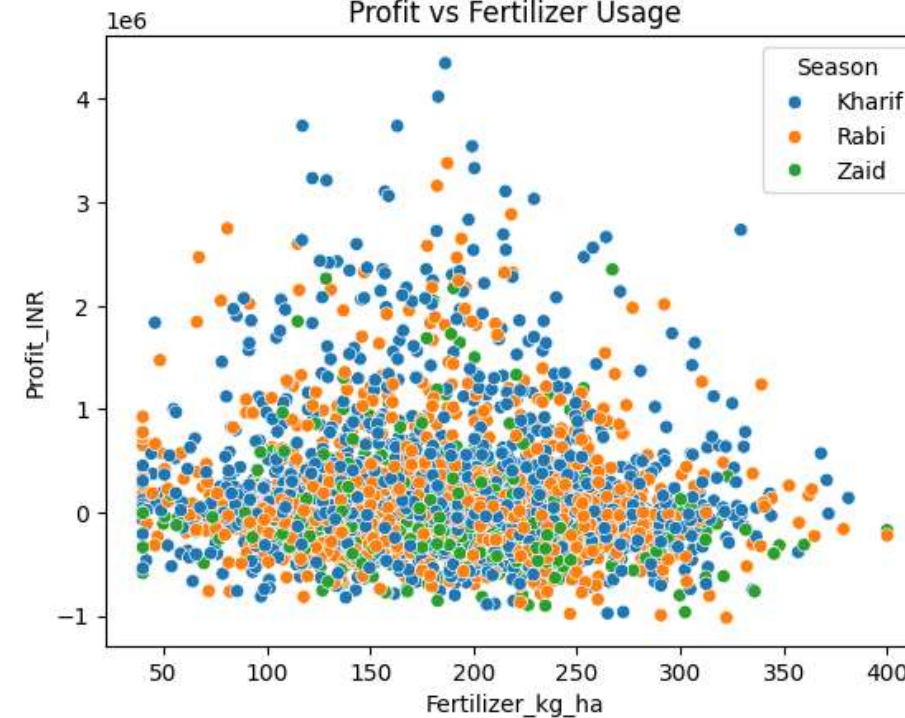
Average Crop Yield per Season



Bivariate: Water vs Yield

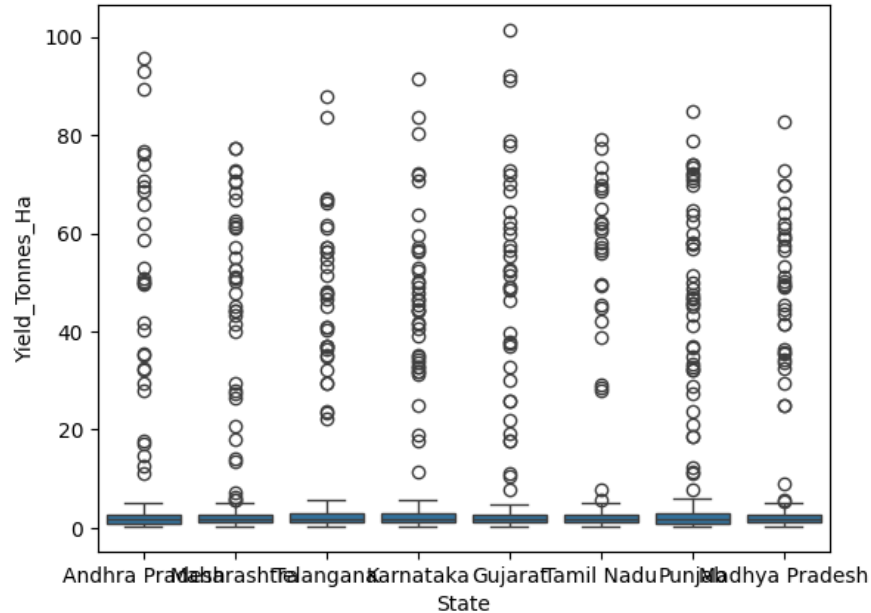


Profit vs Fertilizer Usage

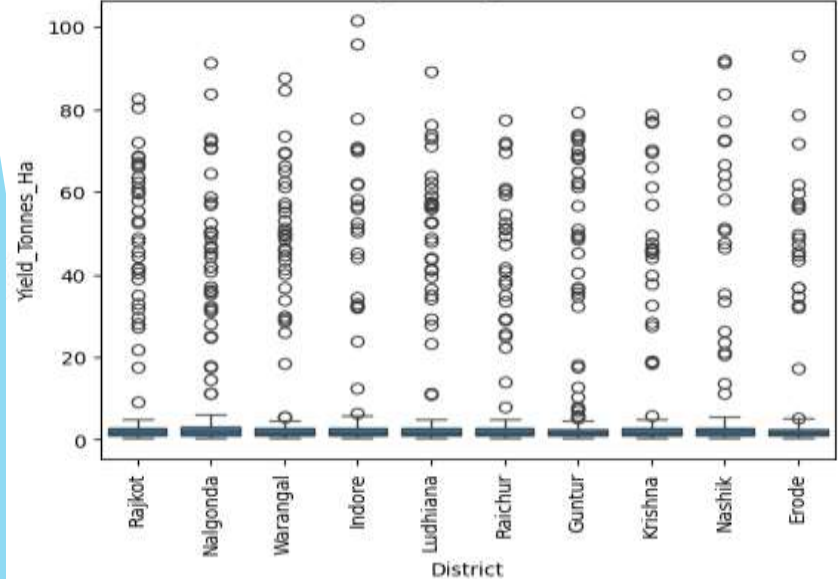


RESULTS

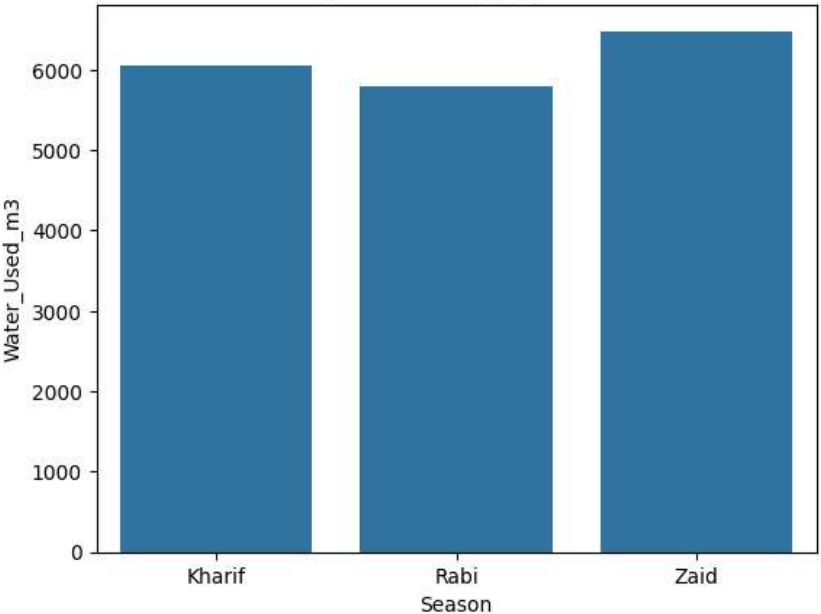
Crop Yield by State



Crop Yield by District



Average Water Usage per Season



Seasonal Performance:					
Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	\
0 Kharif	7.931118	850.658559	28.441153	71.829107	
1 Rabi	7.730558	436.827982	23.491307	57.952982	
2 Zaid	8.124815	299.072935	31.035325	51.964148	
Sunlight_Hours_Day	Soil_ph	Soil_Moisture_pct	Nitrogen_kg_ha	\	
0 6.790086	6.730075	31.199769	120.077003		
1 7.590673	6.674518	24.070241	120.212627		
2 8.179789	6.689244	19.180316	120.307381		
Phosphorus_kg_ha	...	Seed_Quality_Score	Yield_Tonnes_Ha	\	
0 58.044207	...	0.821862	5.673602		
1 57.341815	...	0.831796	4.980895		
2 58.262039	...	0.822777	4.643374		
Production_Tonnes	Market_Price_INR_Tonne	Total_Cost_INR	Revenue_INR	\	
0 46.514951	45048.108357	530950.559654	712489.576369		
1 40.654740	44988.611041	512126.729695	598348.821701		
2 39.341213	43848.179262	545191.304042	518555.615114		
Profit_INR	Water_Used_m3	Water_Efficiency_t_per_1000m3	\		
0 181539.016715	6053.389049	5.929612			
1 86222.092005	5791.909898	5.138018			
2 -26635.688928	6481.217926	4.438399			
Disease_Pest_Risk_pct					
0 54.453545					
1 40.519036					
2 38.105448					

Future scope

- Extend analysis to **multi-year datasets**.
- Integrate **machine learning models** for crop yield prediction.
- Build **interactive dashboards** for farmers.
- Explore **climate change impact** on seasonal agriculture.

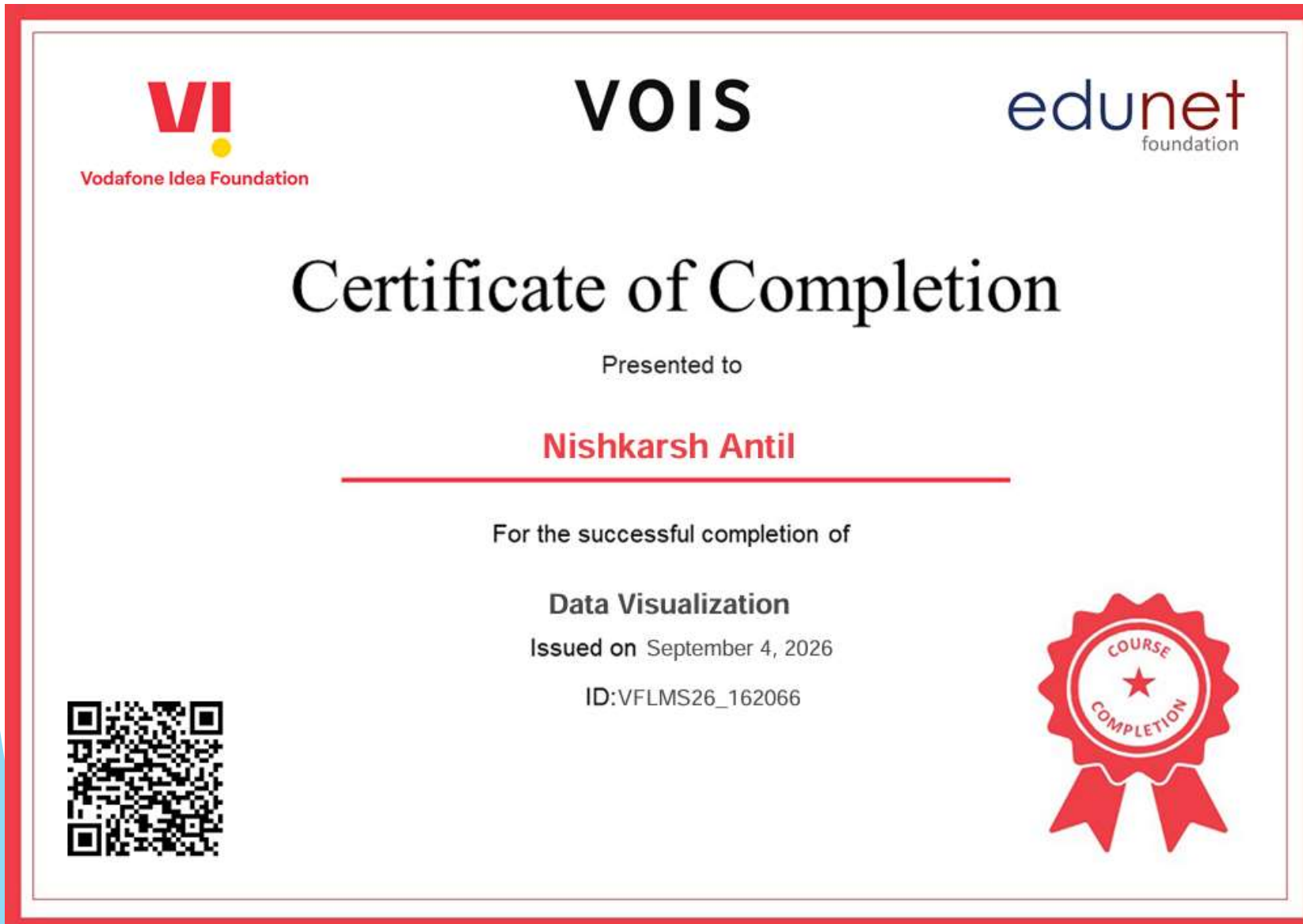


GitHub Link

<https://github.com/nishkarshantil0111-bit/seasonal-agriculture-performance-.git>

VOIS Course completion certificate -

Course name- Data Visualization



Thank you

<https://github.com/nishkarshantil0111-bit/seasonal-agriculture-performance-.git>
AICTE STU ID : STU6a67024e424331785135694